

Ron Huang

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SUMMARY

Senior engineering leader and architect with 20+ years of experience simplifying complex systems into high-impact products. Leading and mentoring cross-functional teams as scrum master and technical lead, shaping a new R&D center and engineering culture while delivering firmware, cloud, and platform solutions. Expert at modernizing mission-critical legacy software and resolving major security issues even with limited source code access. Proven outcomes include 90% faster firmware updates, zero-downtime cloud migrations, and cross-platform solutions from UEFI to VR, combining hands-on engineering with HCI (Human-Computer Interaction) research and award-winning design.

EXPERIENCE

Software Engineer

May 2024 — Present

HP Inc.

Singapore

- Leading one of the inaugural engineering teams in the new R&D center, shaping local engineering culture and practices.
- Serving as scrum master for a newly formed cross-functional team, driving delivery of high-impact platform features.
- Facilitating sprint planning, reviews, and retrospectives to improve team velocity and predictability.
- Mentoring engineers on best practices and continuous improvement within the new R&D organization.
- Delivering firmware for HP PCs that improves stability, security, and user experience for enterprise customers.

Software Engineer

July 2016 — April 2024

HP Inc.

Taipei, Taiwan

- Pioneered 3D video conferencing with real-time stereo reconstruction and auto-framing, elevating remote collaboration.
- Built an extensible biosensor and cognitive-load inferencing platform, enabling new product capabilities.
- Shipped a VR companion app that streamlined controller pairing and privacy setup, reducing onboarding friction.
- Re-architected a test automation framework for cloud-native infra, reducing maintenance while increasing scalability.
- Modernized software packaging with a secure, localized installer, improving user trust and global adoption.
- Led zero-downtime, zero-data-loss migration of legacy applications to AWS, unlocking cloud-native scalability.
- Engineered Thunderbolt firmware update tools that cut deployment time from 30 minutes to 3 minutes.
- Created a SDK that standardized conferencing integrations and accelerated feature delivery across product lines.

Software Engineer

June 2013 — April 2016

Hewlett Packard Enterprise (Hewlett-Packard)

Taipei, Taiwan

- Architected a test automation platform, unifying scheduling, orchestration, and resource management for testing.
- Established DevOps practices that shortened release cycles and improved operational visibility.
- Delivered virtualization test tooling for Hyper-V and vSphere, improving quality coverage across mixed infrastructure.

Staff Engineer

October 2011 — December 2012

Generalplus Technology Inc.

Taipei, Taiwan

- Delivered Android on custom ARM SoCs with a flexible build system, enabling rapid delivery of tailored products.
- Overhauled Android porting process to align with platform standards, shrinking porting time from weeks to days.

Ph.D. Student

July 2009 — June 2011

National University of Singapore

Singapore

- Researched and codified *Software Patterns of User Experience*, producing practical patterns that influenced developer-facing UX guidance and publications.

Advanced Software Engineer

December 2007 — June 2009

Sunplus mMedia Inc.

Taipei, Taiwan

- Built a GUI SDK for digital photo frame platforms, enabling faster UI development and product differentiation.

Software Engineer

February 2005 — December 2007

Compal Communications Inc.

Taipei, Taiwan

- Created a C-based GUI framework and unified in-house/third-party components powering feature phone interfaces.

SKILLS

- **Languages:** C#, Python, C++, TypeScript, PowerShell, Lua, YAML, JSON, XML, HTML, CSS, Bash
- **Tools:** Git, regular expressions, Podman/Docker, CMake, MSBuild, InstallShield, Azure DevOps, AWS CloudFormation, Amazon ECS, Vagrant, Puppet, Jenkins, Makefile, Unity 3D
- **Frameworks / Platforms:** .NET, Win32, WPF, UWP, React, Node.js, OpenGL, ONNX, CUDA, OpenXR, WebRTC

EDUCATION

University of Southern California

M.S. Computer Science

- Area of study: Human-Computer Interaction
- Major GPA: 3.95

December 2004

Los Angeles, California, USA

National Taiwan University

B.S. Electrical Engineering

- Major GPA: 3.73

June 2002

Taipei, Taiwan

AWARDS

Unsung Heroes

HP Inc.

December 2019

Leading the Way

Hewlett-Packard

May 2015

NUS Research Scholarship

National University of Singapore

July 2009

Honorable Mention

Google Desktop Gadget Contest, Google

June 2006

Fifth Place

2nd Annual Computer Society International Design Competition, IEEE

June 2001

PUBLICATIONS

Software Patterns of User Experience: Giving Software Developers More Say in the Design Process

5th Conference on Designing Pleasurable Products and Interfaces

Jun 2011

Milan, Italy

Investigation of Software Patterns of User Experience

7th Conference on Advances in Computer Entertainment Technology

November 2010

Taipei, Taiwan

User experiences and lessons learned from developing and implementing an immersive game for the science classroom

Proceedings of HCI International 2005

July 2005

Las Vegas, Nevada, USA

SIDE PROJECTS (*SELECTED*)

Turnip Calculator (<https://github.com/ronhuang/Turnip-Calculator>)

April 2020

- LINE chatbot that predicts in-game turnip prices for Animal Crossing: New Horizons.
- Enable friends to share forecasts in group chats, becoming a quarantine-era social lifeline.

Where is Makkah (<https://github.com/ronhuang/where-is-makkah>)

July 2011

- Windows Phone 7 app that shows the direction to Makkah (Mecca).

Vistroller (<https://github.com/ronhuang/vistroller>)

May 2011

- Arduino shield that captures inputs from PS2 controller and renders AR markers onto the LCD panel in real-time.
- Android library for translating captured markers into key events.

Grace Birthday (<https://github.com/ronhuang/grace-birthday>)

September 1998

- Windows media gallery in Visual Basic 6, implementing GUI event loops, file I/O, and memory management.